

EXP NO 4	VIRTUAL PRIVATE SERVER (VPS) PROVISIONING AND CONFIGURATION
Date: 05.08.2026	

Aim

To set up a Virtual Private Server (VPS) environment using VirtualBox, install the Ubuntu Server operating system, configure network settings such as IP address, subnet mask, gateway, and DNS, and enable secure remote access using SSH.

What is a VPS?

A Virtual Private Server (VPS) is a virtual server created using virtualization technology. It operates as an independent server with its own operating system, storage, memory, and network configuration, even though the physical resources of the host system may be shared with other virtual machines. Virtualization allows physical computing resources to be utilized efficiently and is an important technology used in cloud computing environments.

Network Configuration Parameters

Parameter	Description
IP Address	A unique numerical identifier assigned to a device on a network. It helps identify the device and enables communication with other systems.
Subnet Mask	Used to determine the network and host portions of an IP address. It helps identify which devices belong to the same network.
Gateway	Acts as a communication point between the local network and external networks, allowing devices to send and receive data outside their local network.
DNS	Converts human-readable domain names into their corresponding IP addresses, helping computers locate websites and other network services.
SSH	A secure communication protocol used to remotely access and manage Linux-based systems through a command-line interface.

Procedure

Step 1: Check Internet Connectivity

The Ubuntu terminal was opened and the internet connection was verified by executing the following command:

```
ping google.com
```

Step 2: Check Network Routing

The current network routing information of the Ubuntu system was displayed using:

```
ip route
```

Step 3: Verify DNS Configuration

The DNS configuration was checked by executing:

```
cat /etc/resolv.conf
```

Step 4: Update Package Repository

The Ubuntu package repository information was updated using:

```
sudo apt update
```

Step 5: Install OpenSSH Server

The OpenSSH Server package was installed to enable secure remote connections:

```
sudo apt install openssh-server -y
```

Step 6: Enable SSH Service

The SSH service was enabled so that it starts automatically whenever the system boots:

```
sudo systemctl enable ssh
```

Step 7: Start SSH Service

The SSH service was started using the following command:

```
sudo systemctl start ssh
```

Step 8: Check SSH Service Status

Whether the SSH service was running correctly was verified by executing:

```
sudo systemctl status ssh
```

The service displayed the status as “active (running)”, confirming it was functioning correctly.

Step 9: Find the IP Address

The IP address assigned to the Ubuntu system was obtained using:

```
hostname -I
```

Step 10: Establish SSH Connection

A secure remote connection to the Ubuntu system was established using:

```
ssh <username>@<IP_address>
```

Here, <username> was replaced with the Ubuntu username and <IP_address> with the IP address obtained in the previous step.

Step 11: Verify SSH Login

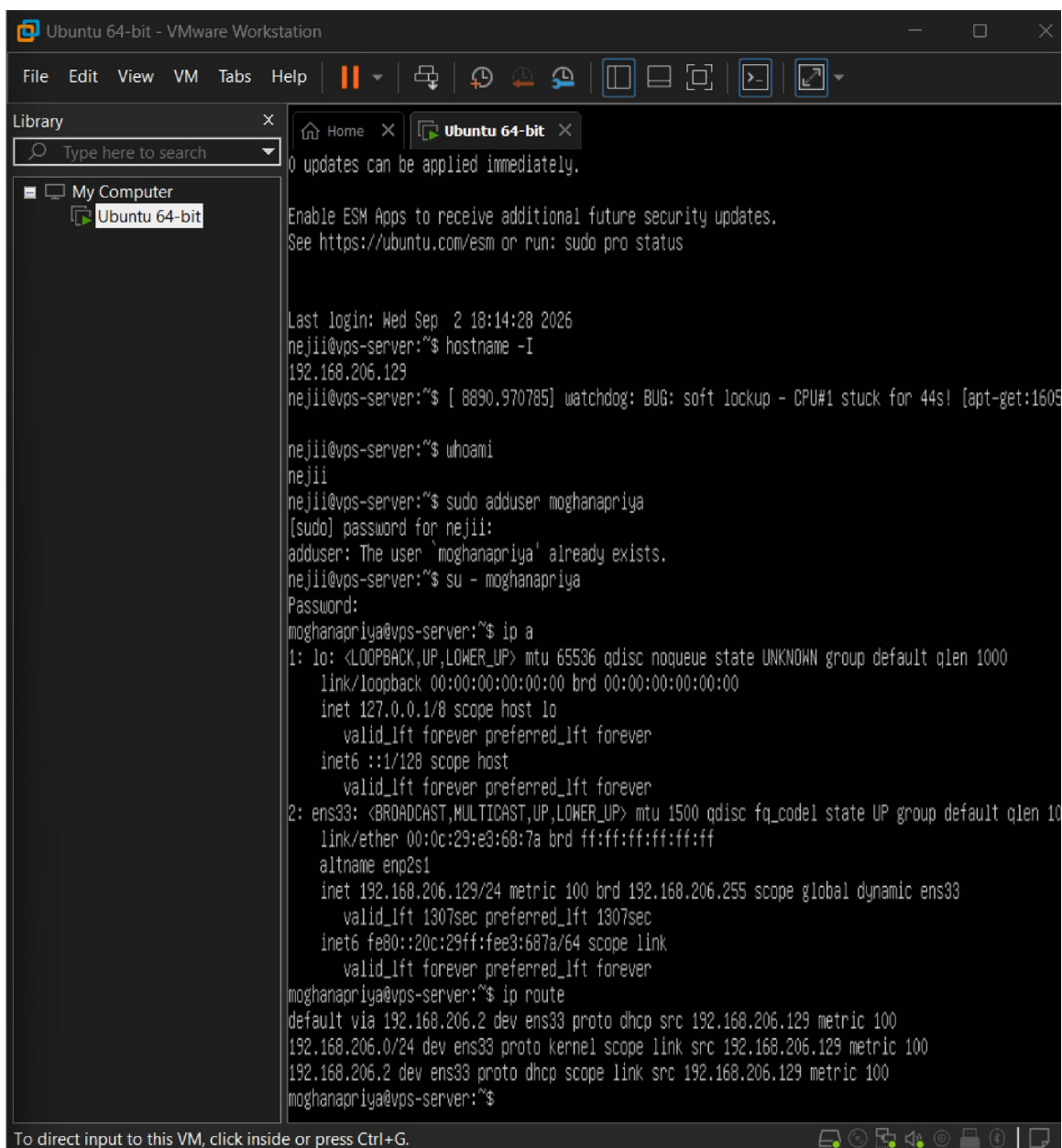
The SSH connection was confirmed to have been successfully established by checking the Ubuntu welcome / login message displayed in the terminal.

Screenshots – VPS Terminal Session

```

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  Ubuntu 64-bit
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
Last login: Wed Sep 2 18:14:28 2026
nejii@vps-server:~$ hostname -I
192.168.206.129
nejii@vps-server:~$ [ 8890.970785] watchdog: BUG: soft lockup - CPU#1 stuck for 44s! [apt-get:1605
nejii@vps-server:~$ whoami
nejii
nejii@vps-server:~$ sudo adduser moghanapriya
[sudo] password for nejii:
adduser: The user `moghanapriya' already exists.
nejii@vps-server:~$ su - moghanapriya
Password:
moghanapriya@vps-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 10
    link/ether 00:0c:29:e3:68:7a brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.206.129/24 metric 100 brd 192.168.206.255 scope global dynamic ens33
        valid_lft 1307sec preferred_lft 1307sec
    inet6 fe80::20c:29ff:fee3:687a/64 scope link
        valid_lft forever preferred_lft forever
moghanapriya@vps-server:~$ _
To direct input to this VM, click inside or press Ctrl+G.

```



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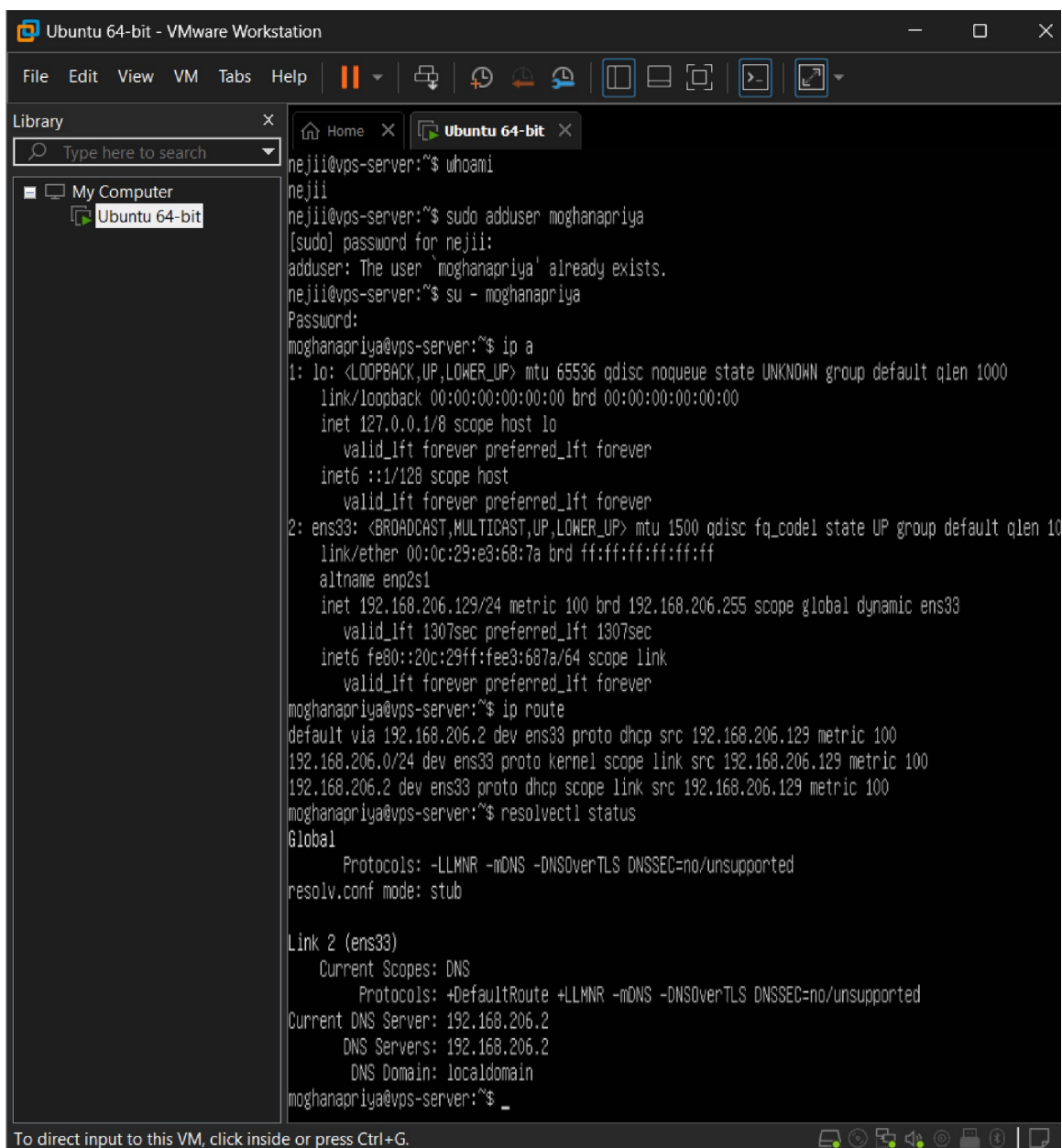
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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

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Password:
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    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:e3:68:7a brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.206.129/24 metric 100 brd 192.168.206.255 scope global dynamic ens33
        valid_lft 1307sec preferred_lft 1307sec
    inet6 fe80::20c:29ff:fe3:687a/64 scope link
        valid_lft forever preferred_lft forever
moghanapriya@vps-server:~$ ip route
default via 192.168.206.2 dev ens33 proto dhcp src 192.168.206.129 metric 100
192.168.206.0/24 dev ens33 proto kernel scope link src 192.168.206.129 metric 100
192.168.206.2 dev ens33 proto dhcp scope link src 192.168.206.129 metric 100
moghanapriya@vps-server:~$
```

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adduser: The user `moghanapriya' already exists.
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192.168.206.0/24 dev ens33 proto kernel scope link src 192.168.206.129 metric 100
192.168.206.2 dev ens33 proto dhcp scope link src 192.168.206.129 metric 100
moghanapriya@vps-server:~$ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 2 (ens33)
    Current Scopes: DNS
        Protocols: +DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    Current DNS Server: 192.168.206.2
    DNS Servers: 192.168.206.2
    DNS Domain: localdomain
moghanapriya@vps-server:~$ _
```

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```

nejii@vps-server:~$ su - moghanapriya
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

moghanapriya@vps-server:~$ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
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Link 2 (ens33)
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    Current DNS Server: 192.168.206.2
    DNS Servers: 192.168.206.2
    DNS Domain: localdomain

moghanapriya@vps-server:~$ sudo systemctl status ssh
[sudo] password for moghanapriya:
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2026-09-02 18:13:28 UTC; 1 week 0 days ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 844 (sshd)
     Tasks: 1 (limit: 4846)
    Memory: 5.4M
       CPU: 73ms
    CGroup: /system.slice/ssh.service
            └─844 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 02 18:13:28 vps-server systemd[1]: Starting OpenBSD Secure Shell server...
Sep 02 18:13:28 vps-server sshd[844]: Server listening on 0.0.0.0 port 22.
Sep 02 18:13:28 vps-server sshd[844]: Server listening on :: port 22.
Sep 02 18:13:28 vps-server systemd[1]: Started OpenBSD Secure Shell server.
Sep 02 18:48:14 vps-server sshd[15686]: Accepted password for nejii from 192.168.206.129 port 4693
Sep 02 18:48:14 vps-server sshd[15686]: pam_unix(sshd:session): session opened for user nejii(uid=
lines 1-18/18 (END)

```

To direct input to this VM, click inside or press Ctrl+G.

```

neji@vps-server:~$ su - moghanapriya
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

moghanapriya@vps-server:~$ resolvectl status
Global
  Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  resolv.conf mode: stub

Link 2 (ens33)
  Current Scopes: DNS
    Protocols: +DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  Current DNS Server: 192.168.206.2
  DNS Servers: 192.168.206.2
  DNS Domain: localdomain
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Sep 02 18:48:14 vps-server sshd[15686]: pam_unix(sshd:session): session opened for user nejii(
lines 1-18/18 (END)

```

Conclusion

A Virtual Private Server environment was successfully set up using VirtualBox and Ubuntu Server. The required network parameters were configured, the OpenSSH server was installed and enabled, and secure remote access to the VPS was established and verified using SSH, thereby meeting the objectives of the experiment.